

P A T E N T A P P L I C A T I O N S P E C I F I C A T I O N

**Constraint-Aware AIoT Power Grid Self-
Healing and Autonomous Fault Recovery
System Using Digital Twin, Multi-Modal
Anomaly Detection, Topology-Aware Fault
Localization, Cascading Risk Prediction, and
Recovery Optimization**

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ABSTRACT OF THE DISCLOSURE

A system and method for autonomous self-healing of electrical power distribution networks comprising: a virtual power grid simulation engine **110** configured to model electrical topology and compute real-time AC power flow; a virtual Internet of Things (IoT) sensor network **120** generating multi-modal electrical measurements including voltage, current, frequency, active power, reactive power, and temperature; a multi-layer anomaly detection module **140** combining rule-based electrical constraint monitoring with isolation forest machine learning; a topology-aware fault localization engine **160** utilizing graph-theoretic analysis of the grid network to identify faulted components; a cascading failure risk predictor **170** that estimates secondary failure probabilities through graph propagation modeling; a constraint-aware recovery optimization engine **180** that generates candidate recovery configurations, validates each against hard electrical constraints including thermal limits, voltage bounds, transformer capacity, network connectivity, and critical load preservation, and selects an optimal feasible recovery plan; and a closed-loop verification module **195** that re-simulates power flow post-recovery and iterates if stability is not achieved. The system operates as a software-based digital twin **199** enabling reproducible experimentation of grid self-healing without physical hardware.

TECHNICAL FIELD OF THE INVENTION

The present invention relates generally to electrical power grid monitoring, fault detection, and autonomous recovery systems. More particularly, the invention relates to an integrated Artificial Intelligence of Things (AIoT) platform that combines virtual sensor infrastructure, multi-modal anomaly detection, topology-aware fault localization, cascading failure prediction, and constraint-aware recovery optimization to enable autonomous self-healing of electrical power distribution networks.

BACKGROUND OF THE INVENTION

Electrical power grids are among the most critical and complex engineered systems in existence, comprising interconnected generators, transformers, transmission lines, substations, and millions of

end-user loads. Maintaining grid stability requires continuous monitoring of electrical parameters and rapid response to faults.

Conventional Supervisory Control and Data Acquisition (SCADA) systems and Energy Management Systems (EMS) provide real-time monitoring of grid parameters. These systems typically employ fixed-threshold alarms that trigger when measurements exceed predefined limits. While effective at detecting gross abnormalities, such systems suffer from significant limitations:

First, traditional monitoring systems detect anomalies but lack the capability to classify fault types, localize fault positions within the grid topology, or predict cascading consequences. An operator receives an alarm indicating abnormal voltage at a particular bus but must manually determine whether the cause is a line failure, transformer overload, generator trip, or sensor malfunction.

Second, existing systems do not model cascading failure propagation. A single component failure can redistribute power flows across the network, overloading adjacent components and triggering sequential trips. The 2003 Northeast Blackout and the 2012 India Blackout demonstrate that cascading failures can escalate from a single local fault to a continental-scale outage within minutes.

Third, recovery actions in conventional systems are either manually determined by human operators or based on pre-programmed lookup tables. Neither approach accounts for the real-time electrical state of the grid. An automated switching action that appears correct in isolation may violate thermal limits on alternate paths, creating new faults while resolving the original one.

Fourth, the proliferation of IoT sensors introduces data reliability challenges. Sensor failures, communication dropouts, and corrupted measurements can produce readings indistinguishable from genuine grid faults. Acting on erroneous sensor data leads to unnecessary and potentially dangerous switching operations.

Fifth, experimentation with grid self-healing algorithms on physical infrastructure is impractical due to safety risks, regulatory constraints, and the inability to reproduce fault conditions. There exists no comprehensive software environment that faithfully simulates the complete detect-diagnose-localize-predict-recover-verify cycle.

SUMMARY OF THE INVENTION

The present invention provides a system and method for autonomous power grid self-healing that addresses the deficiencies of the prior art through an integrated AIoT architecture comprising seven novel subsystems operating in a closed loop.

In one aspect, the invention provides a **multi-modal virtual IoT sensing layer** that generates realistic, temporally correlated electrical measurements from a simulated sensor network, including voltage, current, frequency, active power, reactive power, temperature, power factor, and equipment status, with injected realistic noise, daily load patterns, and fault-induced deviations.

In another aspect, the invention provides a **multi-layer anomaly detection engine** that combines: (a) rule-based electrical constraint monitoring enforcing hard physical limits on voltage, current, frequency, and temperature; (b) machine learning anomaly detection using isolation forest algorithms operating on multi-dimensional feature vectors within rolling time windows; and (c) temporal pattern analysis detecting sudden step changes, gradual degradation, persistent abnormality, and oscillatory behavior.

In yet another aspect, the invention provides a **topology-aware fault localization engine** that represents the electrical grid as a directed graph and uses graph-theoretic analysis combined with sensor correlation data to rank components by fault probability, producing a ranked list of candidate fault locations with confidence scores and affected region identification.

In a further aspect, the invention provides a **cascading failure risk predictor** that, upon fault detection, simulates the redistribution of power flows across the network graph, estimates the probability of secondary component failures, and computes a quantitative cascade risk score that informs recovery planning.

In a still further aspect, the invention provides a **constraint-aware recovery optimization engine** that: (a) generates multiple candidate recovery configurations including component isolation, alternate path routing, load shedding, and generation redistribution; (b) validates each candidate against hard electrical constraints including line thermal capacity, transformer loading limits, voltage bounds, network connectivity, generator capacity, power balance, and critical load preservation; (c) rejects candidates violating any hard constraint; (d) scores feasible candidates using a multi-objective function balancing load restoration, cascade risk minimization, switching cost, and overload penalty; and (e) selects the optimal feasible plan for execution.

In another aspect, the invention provides a **closed-loop verification module** that, after recovery execution, re-runs power flow simulation, collects new virtual sensor measurements, evaluates grid stability, and triggers iterative re-planning if stability criteria are not met.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a formal block diagram of the overall constraint-aware AIoT power grid self-healing architecture showing data flow between subsystems.

FIG. 2 is a detailed flowchart of the closed-loop autonomous self-healing process execution sequence from sensor ingest to verified recovery.

FIG. 3 is an electrical single-line schematic diagram of the virtual power grid topology including power plant, substations, transmission lines, transformers, and critical loads.

FIG. 4 is an architectural schematic of the multi-layer anomaly detection, fault classification, and topology-aware localization subsystem.

FIG. 5 is a process flow diagram of the constraint-aware recovery optimization engine illustrating candidate generation, hard constraint filtering, and multi-objective scoring.

FIG. 6 is a schematic layout diagram of the digital twin human-machine interface and control center dashboard.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

1. Integrated System Architecture (FIG. 1)

Referring to **FIG. 1**, the system **100** comprises a virtual power grid simulation module **110**, a virtual IoT sensor network **120**, a data ingestion and validation layer **130**, a multi-layer anomaly detection engine **140**, a fault classification module **150**, a topology-aware fault localization engine **160**, a cascading failure risk predictor **170**, a constraint-aware recovery optimization engine **180**, an action executor **190**, a post-recovery verification module **195**, and a digital twin visualization interface **199**.

The virtual power grid simulation module **110** computes real-time AC power flow solutions using Newton-Raphson iterations, determining bus voltage magnitudes, phase angles, line active/reactive powers, and equipment utilization percentages.

The virtual IoT sensor network **120** generates continuous multi-modal telemetry comprising voltage (\$V\$), current (\$I\$), frequency (\$f\$), active power (\$P\$), reactive power (\$Q\$), temperature (\$T\$), power factor (\$PF\$), breaker status, and line status.

2. Closed-Loop Self-Healing Process (FIG. 2)

Referring to **FIG. 2**, the self-healing process **200** executes autonomously. Step **201** ingests live telemetry. Step **202** runs rule-based and ML anomaly checks. Step **203** cross-validates sensor channels to reject sensor hardware glitches. Step **204** classifies the fault into one of twelve fault categories. Step **205** computes graph-theoretic component localization. Step **206** calculates severity. Step **207** estimates cascading failure probability on adjacent branches.

Steps **208–213** isolate the faulted branch, synthesize multiple recovery candidates, enforce physical constraint gates, score feasible candidates, and select the optimal plan. Step **214** commands breaker actions. Step **215** re-computes power flow. Steps **216–217** confirm stability or iterate.

CLAIMS

What is claimed is:

1. A system for autonomous self-healing of an electrical power grid, comprising:

- (a) a power grid simulation engine **110** configured to model an electrical network topology comprising buses, generators, transmission lines, transformers, loads, and switching devices, and to compute power flow solutions producing voltage, current, and loading values for each network component;
- (b) a virtual sensor network **120** comprising a plurality of software-simulated sensor agents, each associated with a grid component and configured to generate periodic multi-modal measurements including voltage, current, frequency, active power, reactive power, and temperature;
- (c) a multi-layer anomaly detection module **140** comprising a rule-based electrical constraint checker, a machine learning anomaly detector, and a temporal pattern analyzer, configured to identify abnormal grid conditions from the sensor measurements;
- (d) a fault classification module **150** configured to categorize detected anomalies into a plurality of fault types including transmission line failure, transformer overload, voltage instability, frequency disturbance, generator failure, and sensor malfunction;
- (e) a topology-aware fault localization engine **160** configured to represent the grid as a graph and compute fault probability rankings for grid components based on sensor correlations and network topology;
- (f) a cascading failure risk predictor **170** configured to simulate power flow redistribution following a fault and estimate secondary failure probabilities;
- (g) a constraint-aware recovery optimization engine **180** configured to generate candidate recovery configurations, validate each against hard electrical constraints, reject infeasible candidates, and select an optimal feasible recovery plan; and
- (h) a closed-loop verification module **195** configured to re-simulate power flow after recovery execution and iterate recovery planning if stability criteria are not satisfied.

2. The system of claim 1, wherein the hard electrical constraints comprise line thermal capacity limits, transformer loading limits, bus voltage bounds, network connectivity requirements,

generator capacity limits, power balance equations, and critical load preservation requirements.

3. The system of claim 1, wherein the recovery optimization engine computes an objective function $J(P) = w_1 \cdot \text{RestoredLoad} - w_2 \cdot \text{CascadeRisk} - w_3 \cdot \text{SwitchingCost} - w_4 \cdot \text{OverloadPenalty}$.

4. The system of claim 1, wherein the virtual sensor network generates measurements incorporating baseline operating values, stochastic noise, diurnal load patterns, inter-sensor correlations, and fault-induced deviations proportional to fault severity.

5. The system of claim 1, wherein the multi-layer anomaly detection module distinguishes genuine grid faults from sensor faults by cross-referencing adjacent graph sensors.

6. The system of claim 1, wherein the fault classification module employs a machine learning classifier trained on simulated fault scenarios to output fault type probabilities across at least twelve fault categories.

7. The system of claim 1, wherein the fault localization engine uses graph-theoretic analysis to compute a ranked list of candidate fault locations with associated confidence scores and affected region identification.

8. The system of claim 1, further comprising a digital twin visualization interface configured to render the grid topology, component states, sensor readings, fault locations, recovery actions, and system status in real time through a web-based dashboard with WebSocket-driven updates.

9. The system of claim 1, wherein the recovery optimization engine prioritizes restoration of critical loads including hospitals, emergency services, and communication infrastructure over non-critical loads during power-constrained recovery scenarios.

10. A method for autonomous self-healing of an electrical power grid, comprising the steps of:

- (a) continuously ingesting multi-modal sensor measurements from a virtual IoT sensor network monitoring a simulated electrical grid;
- (b) detecting anomalies using a multi-layer detection engine combining rule-based constraint checking, machine learning anomaly scoring, and temporal pattern analysis;
- (c) classifying the detected anomaly into a fault type using a trained machine learning classifier;
- (d) localizing the fault to a specific grid component using graph-theoretic analysis of the grid topology and sensor correlations;
- (e) predicting cascading failure risk by simulating power flow redistribution across the network graph;
- (f) generating a plurality of candidate recovery configurations;
- (g) validating each candidate against hard electrical constraints and rejecting infeasible candidates;
- (h) selecting an optimal feasible recovery plan using multi-objective optimization;
- (i) executing the selected recovery plan on the grid simulation;
- (j) verifying post-recovery grid stability through re-simulation; and
- (k) iterating steps (f) through (j) if stability is not achieved.

11. The method of claim 10, wherein the entire detect-to-recover cycle completes in under two seconds without human intervention.

12. The method of claim 10, wherein the system operates entirely within a software-based simulation environment without requiring physical electrical hardware or physical IoT sensors.

13. A non-transitory computer-readable medium storing instructions that, when executed by a processor, cause the processor to perform the method of claim 10.

14. The system of claim 1, wherein the machine learning anomaly detector comprises an isolation forest model trained on normal operating data and configured to compute anomaly scores from eight-dimensional feature vectors within rolling time windows.

15. The system of claim 1, wherein the cascading failure risk predictor computes a quantitative risk score between 0 and 1 based on severity-weighted graph propagation and identifies specific components at risk of secondary failure.

16. The system of claim 1, wherein the virtual IoT sensor network simulates communication delays, packet drops, and sensor dropouts, and wherein the system distinguishes data unavailability from physical grid faults.

17. The system of claim 1, wherein candidate recovery configurations generated by the recovery engine include component isolation via breaker commands, load transfer to alternate substations, generation output adjustments, and controlled non-critical load shedding.

18. The system of claim 1, wherein the digital twin visualization interface includes clickable grid components opening real-time metrics, AI confidence indicators, historical logs, and fault risk estimates.

19. The method of claim 10, further comprising logging every fault detection, classification, localization, cascade prediction, candidate plan evaluation, and recovery execution in an audit log.

20. The system of claim 1, wherein the entire platform is deployed on a single computing system without external hardware dependency.

FORMAL DRAWINGS OF THE INVENTION

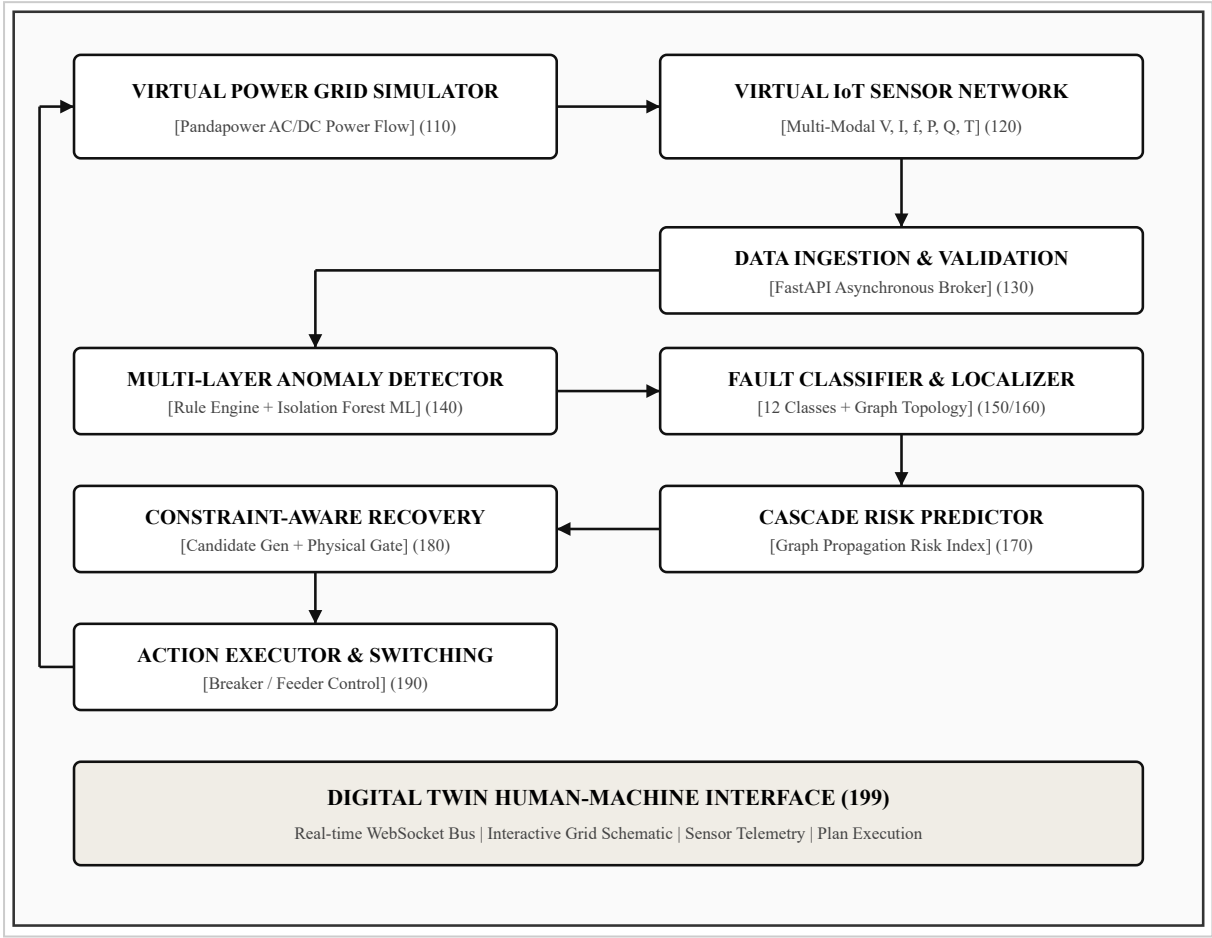


FIG. 1

Overall Constraint-Aware AIoT Power Grid Self-Healing Architecture

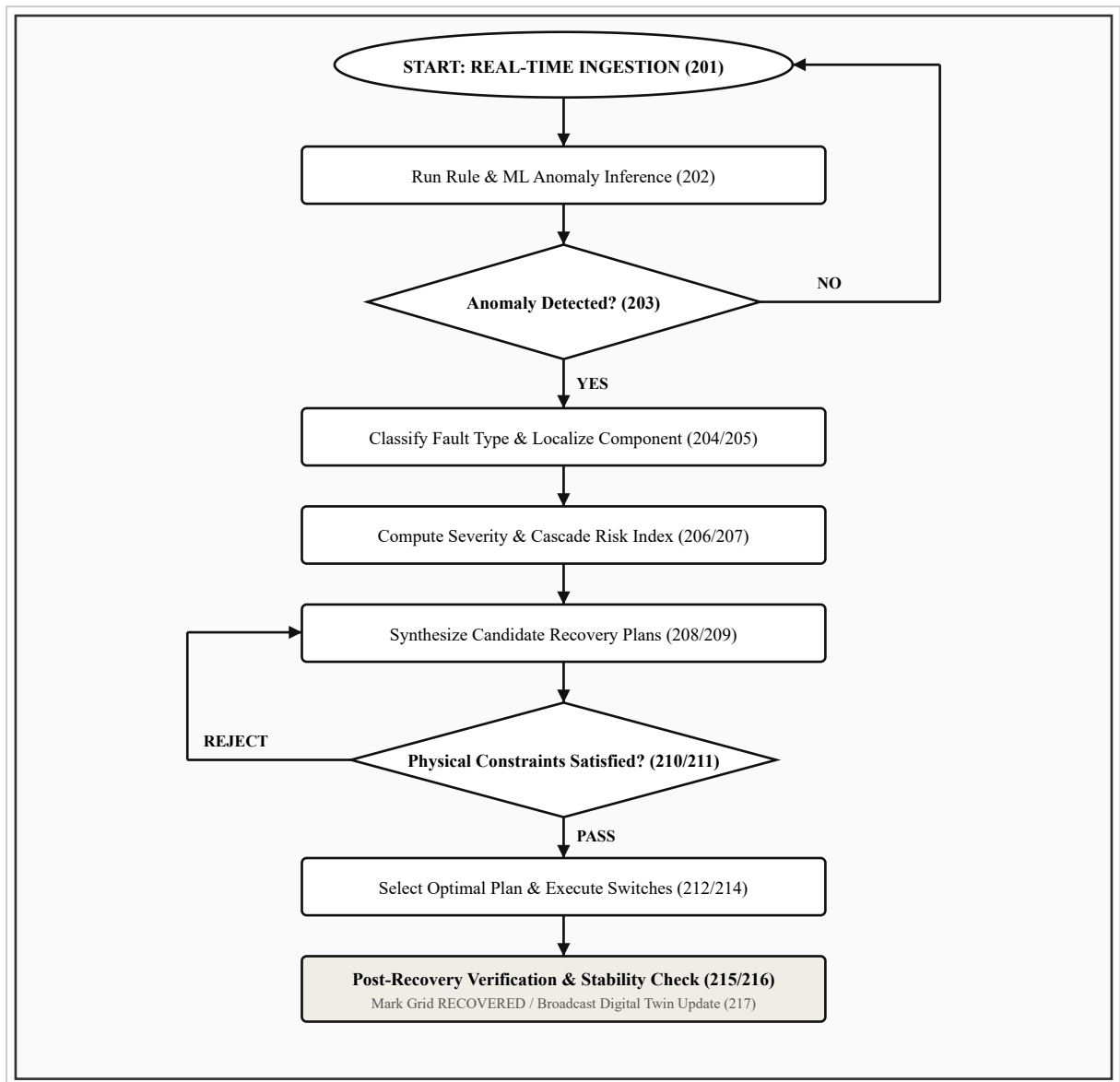


FIG. 2

Flowchart of Autonomous Self-Healing Pipeline

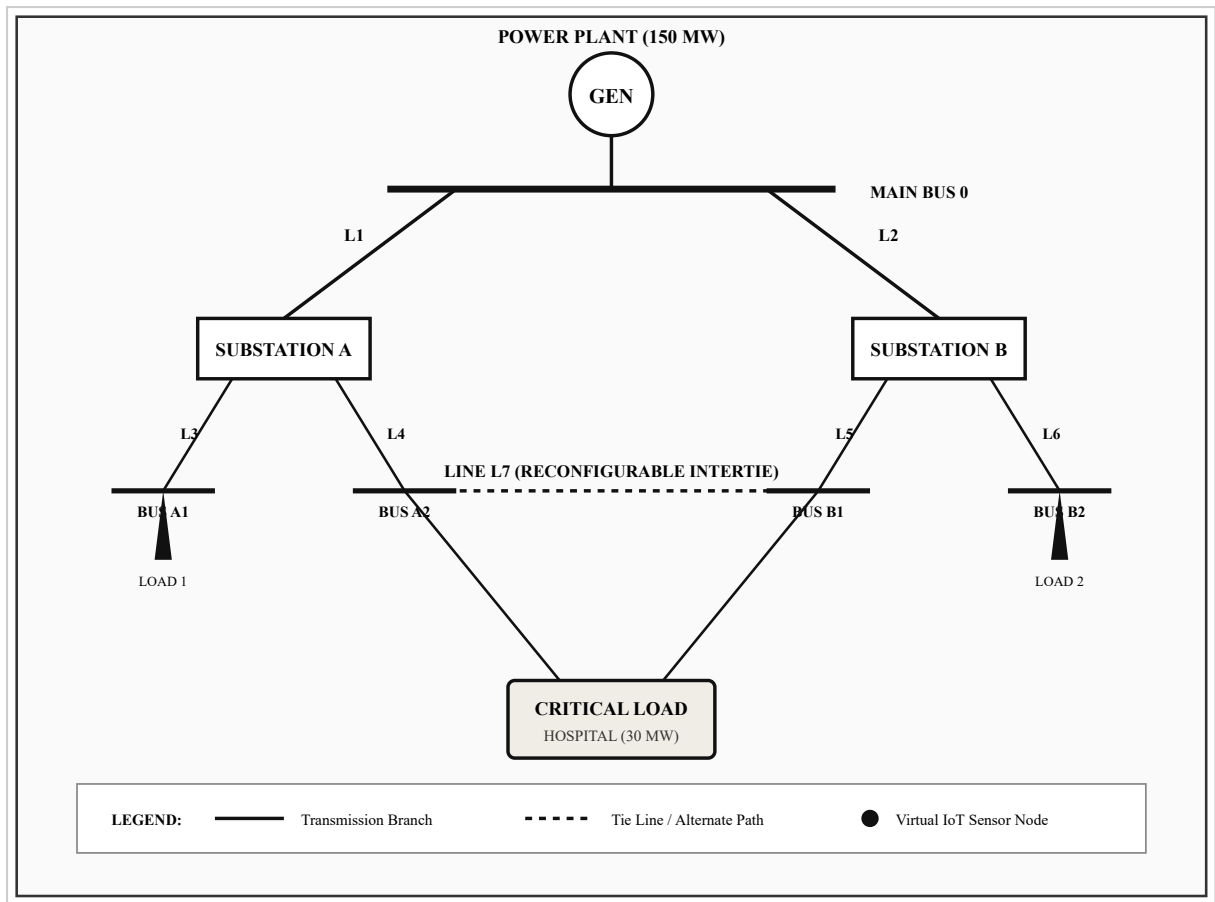


FIG. 3

Virtual Power Grid Single-Line Schematic with Redundant Alternate Routes

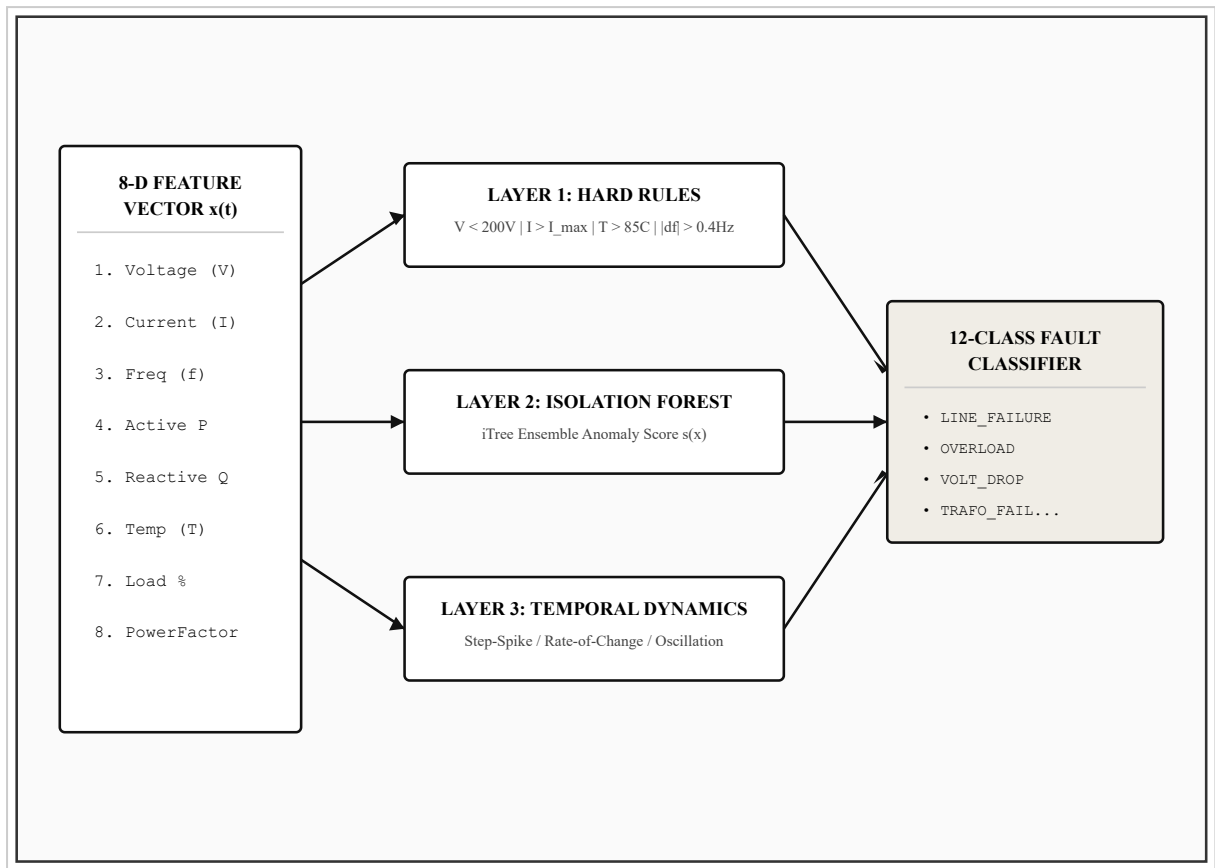


FIG. 4

Multi-Layer Anomaly Detection & Fault Classification Engine Architecture

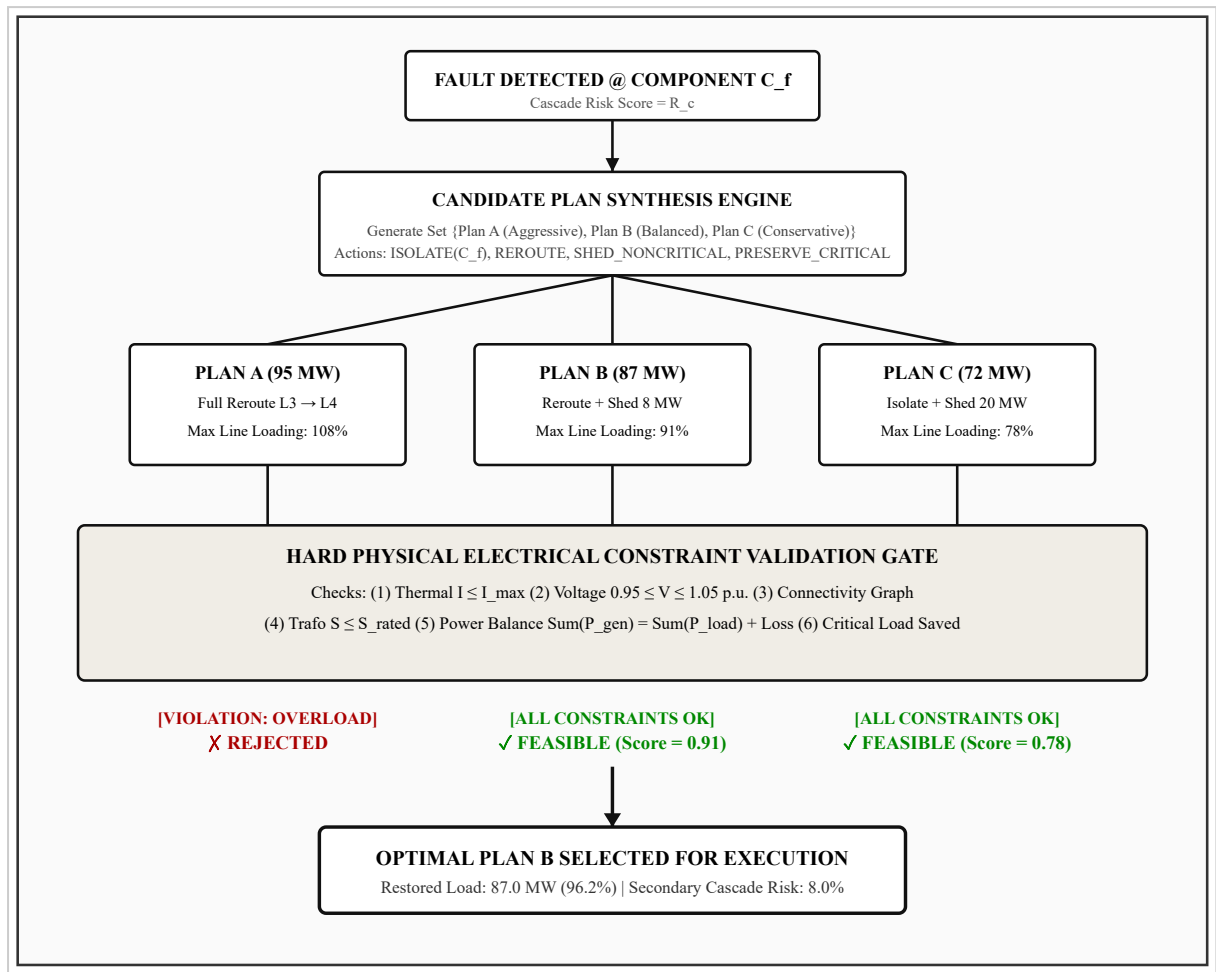


FIG. 5

Constraint-Aware Recovery Optimization Process and Multi-Objective Selection Gate

